BhashaSetu AI

AI-Powered Hindi–Gujarati Communication Assistant

1. Project Overview

BhashaSetu AI is an AI-powered communication assistant designed to reduce language barriers between Hindi and Gujarati speakers.

The application combines speech recognition, multilingual translation, text-to-speech, and intelligent clarification to make everyday and workplace communication easier.

It Integrates Gnani AI speech APIs and Groq LLM to provide language-processing and clarification capabilities.

2. Problem Statement

Language barriers can create difficulties in everyday communication and workplace environments, particularly when people speak different regional languages.

Additionally, workplace instructions may sometimes lack important details, leading to confusion and misunderstandings.

BhashaSetu AI addresses these challenges through two-way Hindi–Gujarati communication and intelligent workplace instruction clarification.

3. Main Features

Feature 1: Hindi to Gujarati Translation

- Accepts Hindi speech input.
- Converts speech into text using speech recognition.
- Generates Gujarati translation.
- Provides Gujarati text-to-speech output.

Feature 2: Gujarati to Hindi Translation

- Accepts Gujarati input.
- Processes the input through the translation pipeline.
- Generates Hindi translation.
- Provides Hindi speech output.

Feature 3: Workplace Clarification

- Processes workplace-related instructions.
- Identifies missing or ambiguous information.
- Generates relevant follow-up questions.
- Helps users provide clearer instructions.

Feature 4: Smart Workplace

- Evaluates workplace instructions.
- Identifies incomplete information.
- Requests clarification when necessary.

- Returns a CLEAR status when the instruction is considered sufficiently detailed.

4. Technologies and APIs Used

Technology| Purpose

Python| Core application development

Gnani Prisma v2.5| Speech-to-text

Gnani Timbre v2.5| Text-to-speech

Gnani Vachana SDK 0.8.1| Integration with Gnani AI services

Groq LLM| Language-model-powered processing and clarification

Pytest| Automated testing

Git and GitHub| Version control and project management

Selected TTS voice: Vikrant

Groq model: “openai/gpt-oss-20b”

5. Gnani AI Integration

Gnani AI speech capabilities are an important part of BhashaSetu AI.

The project uses Gnani Prisma v2.5 for speech recognition and Gnani Timbre v2.5 for speech generation.

These capabilities allow users to interact with the application through spoken language rather than relying only on written text.

6. My Technical Contribution

I worked on designing and developing the BhashaSetu AI application.

My contributions include:

- Developing the Python application structure.
- Integrating Gnani AI speech services.
- Implementing Hindi–Gujarati translation workflows.
- Integrating Groq LLM for intelligent clarification.
- Developing workplace instruction validation and clarification logic.
- Writing automated tests for translation, validation, and clarification.
- Preparing project documentation and managing the codebase using Git and GitHub.

7. Testing and Results

The application was tested using different Hindi and Gujarati inputs and workplace instructions.

Testing covered:

- Hindi-to-Gujarati translation.
- Gujarati-to-Hindi translation.
- Speech processing and audio generation.
- Identification of incomplete workplace instructions.
- Generation of clarification questions.

- Classification of sufficiently detailed instructions.

Automated tests were also implemented to verify important application components.

8. Future Scope

Future improvements may include:

- Support for additional Indian languages.
- More regional language speech capabilities.
- Improved translation quality.
- Additional workplace communication scenarios.
- Better handling of complex and context-dependent instructions.
- A more accessible user interface.

9. Conclusion

BhashaSetu AI demonstrates how AI-powered speech processing, multilingual translation, and intelligent language models can be combined to address practical communication challenges.

The project aims to make communication more accessible between Hindi and Gujarati speakers while helping users create clearer workplace instructions.